Assignment: The Dual-Homed Enterprise Design

Objective: Design a scalable, secure IPv4 addressing scheme for a 500-node organization using Variable Length Subnet Masking (VLSM) and redundant ISP routing.

1. The Scenario

Your client, **Nexus Corp**, occupies a three-story building. They require a robust network for 500 total devices. To ensure "always-on" connectivity, they have contracted both **Globe** and **PLDT** as Internet Service Providers.

The Inventory:

- **Total Hosts:** 500 (includes PCs, printers, and WiFi APs).
- **Departments:**
    1. **Administration:** 50 nodes (High security)
    2. **Engineering:** 200 nodes (High bandwidth)
    3. **Sales & Marketing:** 100 nodes
    4. **Guest WiFi:** 100 nodes (Isolated)
    5. **IT/Server Room:** 50 nodes (Critical infrastructure)

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2. Technical Requirements

Students must apply the following constraints to their design:

- **Private Addressing:** Use the **172.16.0.0/12** or **192.168.0.0/16** private range.
- **Segmentation:** Each department must be on its own **VLAN** to limit broadcast domains (performance) and allow for Firewall ACLs (security).
- **ISP Redundancy:** * **Globe (Primary):** All production traffic.
    - **PLDT (Secondary):** Failover and Guest WiFi traffic.
- **Efficiency:** Use **VLSM** to minimize IP wastage.

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3. Student Task: The Addressing Table

Students must complete a table similar to the one below. (Hint: Remind them to account for the Network ID, Broadcast Address, and Gateway).

Department	Required Hosts	Subnet Mask (CIDR)	Network Address	Usable IP Range
Engineering	200	/24	?	?

Sales	100	/25	?	?
Guest WiFi	100	/25	?	?
Admin	50	/26	?	?
IT/Servers	50	/26	?	?

4. Security & Performance Discussion Points

After the design phase, groups must defend their choices based on these questions:

1. **Security:** How does placing Guest WiFi on a separate subnet protect the Engineering department's intellectual property?
2. **Performance:** Why is a single /23 subnet (512 hosts) worse for performance than five smaller subnets? (Answer: **Broadcast storms**).
3. **Redundancy:** If the Globe router goes down, how will your Gateway address handle the transition to PLDT? (Concepts: **HSRP** or **VRRP**).

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5. Deliverables

- **A Logical Topology Map:** Drawn manually or via Cisco Packet Tracer.
- **IP Implementation Plan:** The completed VLSM table.